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Original Research

Anterior cruciate ligament injury occurrence, return to sport and subsequent injury in the Australian High Performance Sports System: A 5-year retrospective analysis

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ABSTRACT

Objectives: To report anterior cruciate ligament (ACL) injury occurrence, return to sport (RTS) timeframes and ACL subsequent injuries recorded in the Australian High Performance Sports System according to athlete sex.
Methods: ACL injury data of injured athletes were prospectively collected by the treating sports and exercise physicians and physiotherapists within the Australian High Performance Sports System between January 1, 2015 and May 31, 2020. RTS time frames for ACL injuries and the proportion of subsequent ACL injuries were calculated. The RTS time was compared between sexes, age groups and ACL injury categories using the Mann-Whitney *U* test and the Kruskal-Wallis test.
Results: A total of 132 ACL injuries were reported in 108 athletes (77 female, 31 male). ACL injuries accounted for 6.4% of all reported knee injuries, with almost half (48.5%) of the ACL injuries reported to occur during training. Median RTS time was 369 days (IQR = 273–487), with RTS times reducing with increasing age (X^2 (Zbrojkiewicz et al., 2018) = 11.781, p = 0.008). The presence of concurrent knee pathology did not significantly affect the RTS timeframes. ACL injuries were most frequently reported in netball, winter sports, basketball, field hockey and gymnastics. One quarter of the ACL reported (n = 34, 25.8%) were subsequent to a prior ACL injury, with the majority of these injuries occurring to the ipsilateral knee (n = 26) as opposed to contralateral knee (n = 6).
Conclusions: Despite ACL injuries accounting for a small proportion of all knee injuries reported in the Australian High Performance Sports System, more than a quarter are subsequent to a previous ACL injury. Shorter RTS times were observed in older athletes; however, considerable periods of time-loss occur as a consequence of ACL injuries. Effective primary prevention, rehabilitation processes, and ongoing tertiary prevention strategies are warranted in this population.

1. Introduction

Rupture of the anterior cruciate ligament (ACL) can be a career-threatening injury for an athlete (Swenson et al., 2013). The incidence of ACL injuries reported in Australia hospital settings has considerably increased over the past 20 years, increasing 5.2% and 6.2% each year for males and females respectively (Maniar et al., 2022). Australia has one of the highest incidence rates of ACL reconstruction in the world. The rate of ACL reconstructions in Australia increased 43% (from 54.0 to 77.4 per 100 000 population) between 2000 and 2015, with a 74% increase observed in Australians <25 years of age (from 52.6 to 91.4 per 100 000 population) (Zbrojkiewicz et al., 2018). ACL reconstructions and their rehabilitation incur a considerable financial cost for individuals, sporting organisations and the wider health system with an estimated total cost of A$34,079 recently reported in amateur Australian football (soccer) players (Ross, Agresta, et al., 2023).

There are further consequences following ACL injuries, with approximately one in five athletes who return to sport (RTS)

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experiencing a subsequent ACL injury, with 9.9%–11.8% reinjuring their ipsilateral knee and 9.3%–12.2% sustaining a contralateral ACL injury (Akash et al., 2021; Barber-Westin & Noyes, 2020; Everhart et al., 2022). When considering elite and professional level athletes, the rate of ACL reinjury to the ipsilateral knee appears to be lower, with 5.2% suffering graft rupture (Lai, Ardern, et al., 2018). In professional men's soccer 7% of athletes sustained a graft rupture either during the final phase of rehabilitation or within 3 months of RTS, (Waldén et al., 2016) while a second ACL injury rate of 17.8% was found in UEFA soccer players (Villa et al., 2021). Across sport, there does not appear to be a considerable difference in the rate of subsequent ACL injuries according to athlete sex (Akash et al., 2021; Tan et al., 2016). In addition, younger athletes who return to high risk sports also appear susceptible to subsequent ACL occurrence, with 23% sustaining a second ACL injury, of whom approximately 10% sustain ipsilateral graft ruptures and 12% sustain contralateral ACL ruptures (Wiggins et al., 2016). The risk of subsequent injury to either ACL is reported as 30–40 times greater in young athletes following RTS compared to their uninjured counterparts (Wiggins et al., 2016).

The RTS process following ACL injury involves a dynamic continuum that includes an athlete progressing through returning to participation, to preinjury levels of sport and performance (Meredith et al., 2021). Considerable variation has been reported in regards to the participation rate in sport, return to the preinjury level of competition and the length of time taken to RTS following ACL injuries (Ardern et al., 2011; Ross et al., 2020). Prognostic factors, such as age, sex and activity level have established associations with post-injury outcomes (Ardern et al., 2014; Bruder et al., 2023; Glogovac et al., 2019). The presence of concurrent pathology within the knee at the time of ACL surgery is common, (Cristiani et al., 2023; Gupta et al., 2021; Krych et al., 2020; Pedersen et al., 2020; Praz et al., 2019) with inconsistent findings reported in relation to patient outcomes and RTS timeframes. Studies have reported inconsistent associations between meniscal injury at surgery and RTS outcomes.

The Australian High Performance Sport System encompasses a number of sports which have a high published incidence of ACL injuries (e.g. netball, basketball, and field hockey) (Agel et al., 2016; Anderson et al., 2019; Bram et al., 2021; Chia et al., 2022; Downs et al., 2021; Montalvo, Schneider, Webster, et al., 2019). Despite the fact that ACL injuries incur a substantial time, performance and financial burden to athletes and their sporting organisations, (Filbay et al., 2022; Pulici et al., 2023; Ross et al., 2023a, 2023b) an overview of the ACL injuries sustained across Australian High Performance Sports System athletes has not been previously reported. Therefore, the overall aim of this study was to report the first overview of ACL injury occurrence according to athlete sex across the Australian High Performance Sports System. The specific objectives were to determine (a) the frequency of reported ACL injuries, (b) the distribution of timeframes for RTS following an ACL injury, (c) how the presence of concurrent pathology within the knee (e.g., meniscal or chondral injury) affects the timeframe of RTS, and (d) the distribution of subsequent ACL injury following an initial ACL injury for athletes in the Australian High Performance Sports System.

2. Methods

This case series included athletes within the Australian High Performance Sports System who had an ACL injury recorded in the centralised Athlete Management System (AMS) database (Smartabase, Fusion Sport Pty Ltd., Brisbane, Australia) between January 1, 2015 and May 31,2020. The Australian High Performance Sport System has iteratively changed over time, but now comprises of National Sporting Organisations, who receive federal funding through government agencies, as well as state funded and officially endorsed Olympic, Paralympic and Commonwealth Games training centers within the National Institute Network of Australia (Sheehy et al., 2022; Weissensteiner, 2023). A national athlete categorisation framework is used by the Australian High-Performance Sport System to identify talented athletes most likely to achieve performance outcomes linked to Australia's high performance sport targets. Specific athlete categorisation criteria are set by each National Sporting Organisation and the number of athletes categorised at any one time can vary according to funding, the stage of a season and performance outcomes achieved (Australian Sports Commission, 2021). Athletes supported by National Sporting Organisations, or training within the National Institute Network, have a profile on the AMS, which is used as a central repository for athlete data inclusive of injury surveillance data and clinical notes. The number of athletes with an active AMS profile changes throughout the year as developing athletes are added to state and national level squads and other athletes are removed due retirement and deselection.

All injury and medical record data were entered by the treating doctor or physiotherapist into the AMS at the point of care (following initial assessment and throughout rehabilitation). A four-character Orchard Sports Injury Classification System 10 (OSICS-10.1) injury diagnosis code (Rae & Orchard, 2007) was assigned to the injury record. The number of days for which the athlete was unavailable, and days to RTS were calculated. RTS was defined as the date the athlete was cleared by the treating practitioner for full participation in training and competition with no medical restrictions related to the primary mode of training (Australian Institute of Sport, 2020). A subsequent ACL injury was defined as an athlete's second or later ACL injury (whether ipsilateral or contralateral to the initial injury). All definitions and uses comply with the Australian Institute of Sport data dictionary (Australian Institute of Sport, 2020) and the International Olympic Committee consensus statement: methods for recording and reporting of epidemiological data on injury and illness in sport 2020 (Bahr et al., 2020). Ethics approval for the study was granted by the Australian Institute of Sport Human Research Ethics Committee (Approval number: 20200501) and all eligible athletes were contacted via generic email and provided the opportunity to opt-out of the study.

All recorded ACL injuries in the study period were included irrespective of severity (e.g. complete or partial rupture) or management (e.g. surgical or conservative for complete ruptures). Data pertaining ACL injuries (relevant OSICS-10.1 codes: KJAX, KJAP, KJAR, KJAC, KJAG, KUAX, KJBC, KJBX, YKLA) collected during the study period were de-identified (linked by a unique athlete ID) and extracted. Where the code was not unique to ACL injuries (e.g., KJBX refers to multiple ligament injuries), the de-identified first line of treatment notes and the additional diagnosis notes field were reviewed by two authors (JR and LT) to determine if the injury record pertained to an ACL injury.

Independent variables such as the sport, sex, age, injury classification (new or recurrent), when the injury occurred (training, competition or other), side of injury, date of injury, date of surgery (if applicable) and date of RTS were extracted for all ACL injuries. Concurrent OSICS injury diagnosis codes (e.g., co-existing meniscus injury) for each injury record were also extracted. When calculating RTS time, any ACL injury cases that were not followed through until recovery within the AMS (to full training and competition) were right censored – a proxy such as last treatment date for RTS date was not used, therefore these cases were not included in the calculation of RTS time.

Demographic data and injury frequency data were reported using descriptive statistics. As the data retrieval included both current active and archived AMS athlete profiles from the study period, accurate exposure data were not available and therefore it was not possible to accurately determine incidence rates or prevalence of ACL injuries. ACL injuries were compared against all knee injuries reported in the AMS as a proportion, with differences according to athlete sex compared using the Pearson chi squared test. The proportion of athletes who suffered a subsequent ACL injury was calculated. Based on data availability, RTS time frames were calculated for all reported ACL injury types, but RTS rates for this cohort were not able to be calculated. For athletes with more than one ACL injury recorded in AMS, the RTS time frame was also calculated separately to examine if there were differences between RTS

times for the initial ACL and subsequent ACL injuries. Comparison of RTS time between categorical variables (i.e., age group, sex and ACL injury type) was undertaken using the Mann-Whitney *U* test and the Kruskal-Wallis test.

3. Results

A total of 2071 knee injuries (57.8% female, 42.2% males), including surgeries, were recorded in the AMS during the surveillance period. Of these, 6.4% of all knee injuries were ACL injuries (n = 132) (Table 1). The proportion of female athletes with a knee injury who sustained an ACL injury (7.6%) was significantly higher than of male athletes (4.7%) (X^2 (1) = 6.3961, p = 0.011).

One hundred and eight athletes (77 female, 31 male) reported at least one ACL injury in the surveillance period (Table 1). The median age at ACL injury was 21 years for females (IQR 18–25 years) and 20 years (IQR 18–23 years) for males. ACL injuries occurred most frequently in the 18-to-21-year age group, accounting for 33.3% of all ACL injuries (21.9% female athletes, 11.4% male). ACL injuries were reported in athletes from 20 different sports, with ACL injuries most frequently reported in netball (29.5%), winter sports (15.9%) and basketball (12.2%). Training was the activity most frequently (48.5%) participated in at the time of ACL injury (Table 1).

Of the 132 ACL injuries, 56.8% (40.2% female, 16.6% male) were isolated injuries to the ACL and 28.0% (18.2% female, 9.8% male) were ACL injuries with concurrent pathology (Table 1). Of the 37 ACL injuries with concurrent pathology, 18.9% involved both a second ligament and a cartilage related injury.

ACL injuries occurred at an average rate of 2.03 injuries per month. They occurred more frequently in the first half of the calendar year, with the highest rate in May (2.83 ACL injuries per month) (Fig. 1).

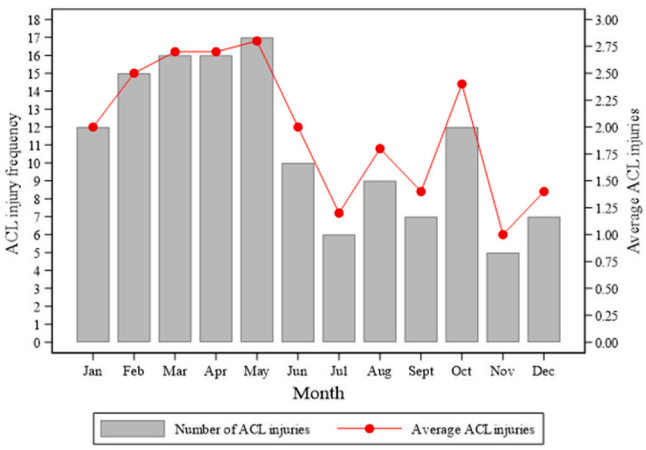
RTS information was available for 77.3% (n = 102) of the records, with a median RTS time of 369 days (IQR 273–487, range 2–822 days) (Table 2). The median RTS time ranged from 344 days (graft rupture) to 402 days (ACL with concurrent injuries) but was considerably lower for partial ACL injuries where the median RTS time was 52 days. Males reported a shorter median RTS time (311 days) than females (373 days), but this did not reach statistical significance (z = 1.042, p = 0.2974). The RTS times became shorter with increasing age – the median RTS time for athletes aged >25 years was 318 days compared to 487 days in athletes aged <18 years (X^2 (3) = 11.781, p = 0.008). The RTS times for athletes with concurrent pathologies (median 402 days, IQR 321–503) were not significantly longer than for injuries involving only the ACL (median 365 days, IQR 263–480) (z = −0.599, p = 0.5493). There was no difference in RTS time between initial (median 377 days, IQR 293–487) and subsequent injuries (median 344 days, IQR 182–503) injuries (z = 1.103, p = 0.2701).

There were 34 subsequent ACL injuries reported; the initial ACL injury in some cases (n = 10 ACLs in 8 athletes) occurred prior to either the athlete's entry into the AMS or the surveillance period, so their initial injury was not included in our results. Noting this, 25.8% of all ACL injuries reported during the study period (n = 34, 20.9% of all injuries in females, and 36.6% of all injuries in males) were subsequent ACL injuries. Subsequent ACL injuries where the initial injury also occurred in the surveillance period represented 18.2% (n = 24) (10.6% female, 7.6% male) of all ACL injuries reported. For the athletes sustaining two or more ACL injuries, the median age at the time of initial injury was 20 years (IQR 18–24 years, median for females 21 years and males 19 years) and at subsequent injury was 22 years (IQR 20–24). At least one subsequent ACL injury was reported in 16.2% of athletes with concurrent pathology at initial injury. The median time between initial and subsequent injury was 404 days (IQR 299–653 days), with 3% of the athletes having not yet been medically classified as returned to full training and competition at the time of their subsequent ACL injury.

Table 1
ACL injuries in the Australian High Performance Sport System from January 1, 2015 to 31/5/2020.

	Female % (n)	Male % (n)	Total % (n)
Knee injuries	57.8 (1196)	42.2 (875)	100.0 (2071)
ACL injuries	68.9 (91)	31.1 (41)	100.0 (132)
Athletes	71.3 (77)	28.7 (31)	100.0 (108)
Number of ACL injuries per athlete			
1 ACL injury	62.0 (67)	21.3 (23)	83.3 (90)
2 ACL injuries	6.5 (7)	5.6 (6)	12.1 (13)
≥3 ACL injuries	*	*	4.6 (5)
Age at time of injury			
< 18	14.4 (19)	5.3 (7)	19.7 (26)
18–21	21.9 (29)	11.4 (15)	33.3 (44)
22–25	16.7 (22)	7.6 (10)	24.3 (32)
> 25	15.9 (21)	6.8 (9)	22.7 (30)
Age statistics			
Median age (years)	21	20	21
IQR (years)	18–25	18–23	18–24
Mean (sd)	21.9 (5.02)	22.2 (7.17)	22.0 (5.7)
Range (years)	15–40	15–47	15–47
ACL injury by sport			
Netball	29.5 (39)	–	29.5 (39)
Winter sports	11.4 (15)	4.5 (6)	15.9 (21)
Basketball	6.1 (8)	6.1 (8)	12.2 (16)
Field Hockey	*	*	10.6 (14)
Gymnastics	*	*	10.6 (14)
Other	9.8 (13)	11.4 (15)	21.2 (28)
ACL injury occurrence			
Training	34.1 (45)	14.4 (19)	48.5 (64)
Competition	28.8 (38)	12.1 (16)	40.9 (54)
Other/Not specified	6.1 (8)	4.5 (6)	10.6 (14)
ACL injury type			
Isolated ACL	40.2 (53)	16.6 (22)	56.8 (75)
ACL with concurrent Injury	18.2 (24)	9.8 (13)	28.0 (37)
Graft Rupture	*	*	6.8 (9)
Chronic ACL Insufficiency	*	*	4.6 (6)
Partial ACL tear	*	*	3.8 (5)
ACL with concurrent injury**			
Second ligament	10.6 (14)	3.8 (5)	14.4 (19)
Cartilage Injury	12.1 (16)	6.1 (8)	18.2 (24)
Other	*	–	*
Subsequent injuries	14.4 (19)	11.4 (15)	25.8 (34)
Number of Athletes	13.9 (15)	10.2 (11)	24.1 (26)
Age at initial injury (median, years) ***	21	19	20
Ipsilateral recurrence	10.6 (14)	9.1 (12)	19.7 (26)
Contralateral injury	*	*	6.1 (8)
Concurrent Pathology at initial injury	*	*	16.2 (6)

ACL = anterior cruciate ligament; n = number IQR = interquartile range; sd = standard deviation.
*Injury count is < 5.
**Note: some athletes had both second ligament and cartilage injuries, and are counted in both rows.
*** Note: 18 athletes sustained initial and subsequent injuries in the surveillance period.



Note: Jan-May = 6 years surveillance; June – Dec = 5 years surveillance

Fig. 1. The frequency and average ACL injury rate in the Australian High Performance Sport System per calendar month from January 1, 2015 to 31/5/2020, inclusive.
Note: Jan–May = 6 years surveillance; June–Dec = 5 years surveillance.

4. Discussion

This is the first study to report ACL injury frequency, RTS timeframes, impact of concurrent knee pathology on RTS timeframes and the proportion and distribution of subsequent ACL injuries in Australian High Performance Sports System athletes. One out of sixteen knee injuries reported were ACL injuries, occurring at an average rate of approximately two injuries per month. Two out of three ACL injuries were in female athletes and half were reported in athletes aged 21 or younger. In most cases, athletes RTS at 12 months post-injury but some variations in RTS time frames were noted between sexes and age groups, with reduced RTS times observed with increasing age. One in six athletes sustained at least one subsequent ACL injury. The presence of concurrent pathology was not associated with RTS outcomes in this athlete population.

Two out of three ACL injuries reported were in female athletes, and a higher proportion of female athletes with a knee injury sustained an ACL injury compared to male athletes. Anatomical (e.g., decreased femoral notch width), physiological (e.g., increased joint laxity), hormonal (e.g., the effect of oestrogen on ACL injury incidence) and biomechanical (e.g., variations in kinematics during cutting maneuvers) factors are all proposed to increase the risk of ACL injuries in female athletes despite contrasting findings across the literature (Devana et al., 2022; Gans et al., 2018; Hewett et al., 2006; Shelbourne et al., 1998; Söderman et al., 2001; Sutton & Bullock, 2013; Tan et al., 2016; Webster, 2021). Socio-cultural factors such as between-gender differences in access to development pathways, high performance training facilities, coaching staff and performance support practitioners, and athlete income also likely contribute to the established higher rates of ACL injury in female athletes (Bruder et al., 2023; Parsons et al., 2021). Limitations of our data made direct comparisons of injury incidence rates between sexes not possible, however our results support previous findings that female athletes represent a key target group for primary ACL injury prevention strategies (Mattu et al., 2022; Montalvo, Schneider, Yut, et al., 2019; Petushek et al., 2019; Sugimoto et al., 2015).

More than half of the ACL injuries were in athletes aged between 16 and 21 years. Given that ACL injury primary prevention initiatives have shown evidence of success in some sports, (Bizzini & Dvorak, 2015; Crossley et al., 2020; Huang et al., 2020; Michaelidis & Koumantakis, 2014) young athletes also represent a key target for ACL injury prevention measures (Ardern et al., 2018; Arundale et al., 2023; Myer et al., 2013; Rössler et al., 2014; Soomro et al., 2016). This data highlights the need to ensure that athletes entering the system are physically and psychologically ready for the level of training they will encounter, and that coaches and performance staff are willing to adjust training to match their athlete's capabilities.

During the surveillance period, ACL injuries occurred at an average

Table 2
Times to RTS following ACL injury in Australian High Performance Sport System athletes from January 1, 2015 to 31/5/2020.

	Number of injuries	Median days to RTS	IQR	Mean (sd) days to RTS	Minimum days to RTS	Maximum days to RTS
All ACL injuries	132	369	273–487	364 (181)	2	822
Gender						
Female	91	373	320–503	380 (179)	2	822
Male	41	311	155–485	313 (185)	37	592
Age Group						
<18	26	487	403–509	445 (141)	37	591
Female	19	487	403–507	469 (81)	355	591
Male	7	496	260–523	391 (237)	37	536
18 - 21	44	383	293–503	386 (160)	10	711
Female	29	402	334–529	421 (160)	10	711
Male	15	311	191–442	328 (157)	149	592
22 - 25	32	349	147–415	321 (217)	26	822
Female	22	358	269–415	349 (221)	26	822
Male	10	154	97–311	204 (179)	46	460
>25	30	318	257–352	262 (138)	2	376
Female	21	318	257–352	262 (138)	2	376
Male*	9	.	.	.	.	.
Injury Type						
Isolated ACL injury	75	373	316–483	399 (156)	147	822
ACL with concurrent injury	37	402	321–503	372 (169)	10	585
Graft rupture	9	344	57–495	298 (220)	36	560
Chronic ACL insufficiency	6	367	2–509	293 (262)	2	509
Partial ACL tear	5	52	37–99	63 (32)	37	99
ACL with Concurrent Injury						
Second ligament	19	403	273–507	376 (188)	10	560
Cartilage Injury	24	402	321–507	382 (162)	26	585
Subsequent ACL Injuries	34	344	182–503	333 (188)	36	720
Ipsilateral injury	26	344	147–509	333 (213)	36	720
Contralateral injury	8	349	320–358	372 (91)	302	529

ACL = anterior cruciate ligament; IQR = interquartile range; sd = standard deviation.
*RTS data was not available/incomplete (count <5 of complete cases).

rate of 2.03 injuries per month, with the rate highest during the first half of the calendar year and peaking in May (2.83 ACL injuries per month), which corresponds with early to mid-season phases for sports like netball and hockey in Australia. Netball, winter sports and basketball had the highest frequency of ACL injuries, reflecting previously reported results, (Agel et al., 2016; Hootman et al., 2007) and emphasizing the need for effective ACL injury prevention strategies in these sports.

Nearly half of all ACL injuries (49.5% female, 46.3% male) occurred in training, in contrast to previous studies that report higher ACL injury incidence rates during competition (Gans et al., 2018; Waldén et al., 2016). Due to a lack of exposure data determining incidence rates during training or competition for our cohort was not possible. Athletes are often added to the AMS for training camps or as part of a broad high-performance squad but may not actually compete while part of the Australian High Performance Sports System. Thus, the ratio of training exposure to competition exposure in our cohort is likely higher than for a professional sporting team where athletes within a small squad compete on a regular basis.

The median time to RTS in this cohort was 369 days. Given the high rates of ACL reconstruction in Australia (Zbrojkiewicz et al., 2018) it is likely that most complete ACL injuries were managed with ACL reconstruction. We were unable to confidently determine the precise rate of ACL reconstruction in our cohort due to limitations of our data, but RTS timeframes for this cohort concur with the timeframes for RTS from ACL reconstruction for other elite sport environments (Lai et al., 2018a, 2018b; Waldén et al., 2016).

We found that the median RTS time for athletes aged >25 years was 169 days faster than that of athletes aged <18 years. To our knowledge no research has previously reported or examined this effect, and the nature of this retrospective study precludes us from knowing the exact reasons for it. Possible contributors are a lower risk tolerance of performance and medical staff, coaches, parents and emerging athletes when planning rehabilitation of young athletes; higher technical ability and physical capacity in more experienced athletes; and pressures on more experienced athletes to be prepared for major events (e.g., Olympic Games). We observed a median RTS time that was 62 days faster for male athletes than female athletes, but this difference was not statistically significant. To our knowledge, there is no available data for sex difference in RTS timeframes in high-performance athletes; this evidence gap is also highlighted in a recent systematic review (Lai, Ardern, et al., 2018). Differences in RTS times for male and female athletes are likely influenced by the same genetic and socio-cultural differences that contribute to the differences in the ACL incidence rate (Bruder et al., 2023; Parsons et al., 2021).

We did not observe any difference between RTS times for initial and subsequent ACL injuries, extending previous research which reported a non-significant 2.2 month increase in RTS timeframes after a subsequent ipsilateral ACL injury in the Australian Football League (Lai, Feller, & Webster, 2018). This may indicate that approximately a year is sufficient time for elite athletes to meet the RTS milestones set by their performance teams. We found 27.8% of athletes who sustained a second ACL injury also sustained a third in the surveillance period. It is possible that for some athletes achieving current RTS milestones may be a poor predictor of subsequent injury risk; that approximately a year of rehabilitation following ACL injury is not sufficient to mitigate an increased risk for subsequent injury in some athletes; or that other factors contribute to greater susceptibility for both initial and subsequent ACL injury.

The median RTS times for ACL injured athletes with a concurrent knee pathology was found to be comparable to athletes with isolated ACL injuries, despite injuries with concurrent pathology often requiring modifications to the rehabilitation process. Of the athletes who sustained an ACL injury with concurrent pathology, 16.2% suffered at least one further ACL injury, almost equal to the subsequent injury proportion of the whole cohort (16.7%). Our data indicates concurrent knee pathology does not affect the RTS time or subsequent injury prognosis of Australian High Performance Sports System athletes with an ACL injury who are successful in returning to sport. This may represent valuable prognostic information to coaches, athletes and medical support staff faced with serious multi-structure knee injuries.

The proportion of ACL injuries reported as a subsequent ACL injury was 25.8%, which is comparable to previous research identifying that approximately 21–25% of young athletes (<25 years) returning to high-risk sports following ACL reconstruction suffer a subsequent ACL injury (Wiggins et al., 2016). Given this cohort consisted of athletes of similar age, with the majority returning to high-risk sports, it is unsurprising to see this similarity. Approximately one in three (36.6%) injuries in male athletes and one in five (20.9%) injuries in female athlete were subsequent ACL injuries. The reasons for the higher secondary injury proportion in male athletes are unclear, but it could be hypothesised that a higher proportion of ACL injured male athletes return to elite level training and competition following primary ACL injury, or that the shorter RTS timeframes of male athletes predispose them to secondary injury.

Our data represents a five-and-a-half year review of ACL injuries in high-performance athletes from across the spectrum of Olympic, Paralympic and Commonwealth games sports. Given that all Australian High Performance Sports System sports utilise the AMS for clinical records, it is likely that relatively few ACL injuries were not reported. A further strength is that each injury was prospectively entered into the AMS by the treating clinician and assigned an OSICS-10.1 code at the time of assessment. However, the AMS is not universally used within the Australian High Performance Sports System as a repository for exposure information meaning we were unable to accurately calculate ACL injury incidence rates. This prevents accurate comparison of risk between athlete sex and sport, which future research should address. To ensure consistency of reporting, a data dictionary (Australian Institute of Sport, 2020) is available to all AMS users in the Australian High Performance Sports System, though ensuring consistent data entry across a widespread system is difficult. Some data was incomplete as not all athletes were followed to RTS within the AMS, and as a result we were unable to calculate RTS rates for this cohort. Furthermore, the management (conservative/surgical) of ACL injuries was not able to be reported as this is not captured as a structured data point within the surveillance system, and the ethical approval granted for this study did not extend to reviewing all the associated clinical management notes linked to the structured injury surveillance fields. The breadth of sports and age ranges of athletes included in this research may enable extrapolation of our findings to other multi-sport high-performance environments, however, inference made to specific sport cohorts should occur with caution given the heterogeneity of athletes and likely risk factors that exist within multisport athlete cohorts.

5. Conclusion

ACL injuries result in considerable time loss from sport in the Australian High Performance Sport System, with one in four athletes reported to experience at least one further subsequent ACL injury. A higher proportion of knee injuries were ACLs in female athletes compared to male athletes, and a reduction in time to RTS was observed with increasing athlete age. Effective primary prevention, rehabilitation processes, and ongoing tertiary prevention strategies that mitigate against subsequent ACL injuries are warranted in this Australian athlete population.

Ethical declaration

Ethics approval for the study was granted by the Australian Institute of Sport Human Research Ethics Committee (Approval number: 20200501).

All eligible athletes were contacted via generic email and provided the opportunity to opt-out of the study.

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Declaration of competing interest

None declared.

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